Math 602: Real analysis II

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Abstract

This is a graduate course on functional analysis, taught by Professor Mark Rudelson. Topics include basic functional analysis, the space $C(K)$, Hilbert space, Fourier series, Fourier integrals, Sobolev spaces, Hausdorff measure and Hausdorff dimension.

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Chapter 1

Normed and Banach Spaces

Lecture 1: Normed and Banach Spaces

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1. Basic functional analysis.
2. The space $C(K)$ (continuous functions on compact metric space K).
3. Hilbert space.
4. Fourier series, Fourier integrals.
5. Sobolev spaces.
6. Hausdorff measure and Hausdorff dimension.

1.1 Normed and Banach Spaces

Definition 1.1.1. Let X be a linear space over $\mathbb{R}$ or $\mathbb{C}$. A function $p : X \to [0, \infty)$ is called a *norm* on X if it satisfies the following properties:

1. $p(x) = 0$ if and only if $x = 0$.
2. For any $\alpha \in \mathbb{R}$ or $\mathbb{C}$ and $x \in X$, we have $p(\alpha x) = |\alpha| p(x)$.
3. For any $x, y \in X$, we have $p(x + y) \leq p(x) + p(y)$.

The pair (X, p) is called a *normed space*.

Example. Take $X = \mathbb{C}$, and $p(x) = |x|$. Then it is clear that X is a normed space.

From now on, for $x \in X$, we use $\|x\|_X$ to denote the norm $p(x)$.

Example. Let $p \in [1, \infty]$, and let $(\Omega, \mathcal{A}, \mu)$ be a measure space. Let $X = L^p(\Omega, \mathcal{A}, \mu)$ be the space of measurable functions $f : \Omega \to \mathbb{C}$, modulo the equivalence relation $f \sim g$ if $f = g$ μ-almost everywhere, such that the L_p norm defined as follows in finite:

$$\|f\|_p = \begin{cases} \left(\int_\Omega |f|^p \, d\mu \right)^{1/p}, & 1 \leq p < \infty, \\ \inf \left\{ t > 0 : \mu\big(\{ x \in \Omega : |f(x)| > t \}\big) = 0 \right\}, & p = \infty. \end{cases}$$

If we take $\Omega = \mathbb{N}$, and define $\mu(\{n\}) = 1$ for $n \in \mathbb{N}$, $\mathcal{A} = 2^{\mathbb{N}}$. In this case $L^p(\Omega, \mathcal{A}, \mu)$ is denoted by ℓ_p with norm

$$\|\{x_n\}_{n=1}^{\infty}\|_p = (\sum_{n=1}^{\infty} |x_n|^p)^{1/p}, \quad 1 \leq p < \infty,$$

and

$$\|\{x_n\}_{n=1}^{\infty}\|_{\infty} = \sup_{n\in\mathbb{N}} |x_n|.$$

There are some interesting subspaces of ℓ_p:

1. $C = \{\{x_n\}_{n\in\mathbb{N}} \in \ell_\infty : \lim_{n\to\infty} x_n \text{ exists}\}$
2. $C_0 = \{\{x_n\}_{n\in\mathbb{N}} \in \ell_\infty : \lim_{n\to\infty} x_n = 0\}$

Example. Let (K, d) be a compact metric space, and let $C(K) = \{f : f \to \mathbb{C}, f \text{ continuous}\}$. Then,

$$\|f\|_\infty = \|f\|_{C(K)} = \max_{x\in K} |f(x)|$$

makes it into a normed space.

Definition 1.1.2. A complete normed space is called a *Banach space*.

Example.

1. For the previous example where we take $\Omega = \mathbb{N}$ and norm ℓ_p, we show that this is a Banach space. Let $x^{(m)} = \{x_n^{(m)}\}_{n=1}^{\infty}$ be a Cauchy sequence in ℓ_p. Notice that we have $\left|x_n^{(m)} - x_n^{(k)}\right| \leq \left\|x^{(m)} - x^{(k)}\right\|_p$. We thus see that each sequence $\{x_n^{(m)}\}_{m=1}^{\infty}$ is a Cauchy sequence. Since $\mathbb{R}$ and $\mathbb{C}$ are complete, each limit exists. Let x_n be the limit of $\{x_n^{(m)}\}_{m=1}^{\infty}$. We claim that $\{x_n\}_{n=1}^{\infty}$ is the limit of $\{x^{(m)}\}_{m=1}^{\infty}$. First, we need to show that $\{x_n\}_{n=1}^{\infty} \in \ell_p$. Since $\{x^{(m)}\}_{m=1}^{\infty}$ is a Cauchy sequence, for any $\varepsilon > 0$, there exists $M > 0$ such that $\left\|x^{(a)} - x^{(b)}\right\|_p < \varepsilon$ for any $a, b > M$. Since $x_n^{(k)} \to x_n$ as $k \to \infty$, we see by Fatous lemma that
$$\sum_{n=1}^{\infty} \left|x_n^{(m)} - x_n\right|^p \leq \liminf_{k\to\infty} \sum_{n=1}^{\infty} \left|x_n^{(m)} - x_n^{(k)}\right|^p \leq \varepsilon^p.$$
We see that $\left\|x^{(m)} - x\right\|_p \leq \epsilon$. Therefore $x^{(m)} - x \in \ell_p$. Thus $x \in \ell_p$. Moreover, $x^{(m)} \to x$ as $m \to \infty$.

2. $C(K)$ is a Banach space with the norm $\|\cdot\|_\infty$. Let $\{f_n\}_{n=1}^{\infty}$ be a Cauchy sequence. For any $x \in K$, $\{f_n(x)\}_{n=1}^{\infty}$ is also a Cauchy sequence. Let $f(x)$ be the limit of $\{f_n(x)\}_{n=1}^{\infty}$, and let f be such a function. We claim that $f_n \to f$ in the norm $\|\cdot\|_\infty$. Choose N such that for any $m, n > N$, $\|f_m - f_n\|_\infty < \varepsilon$. Then, $|f_m(x) - f_n(x)| \leq \varepsilon$ for any $x \in K$. Letting $n \to \infty$, we see that $|f_m(x) - f(x)| \leq \varepsilon$ for any $x \in K$. Therefore, $\|f_m - f\|_\infty \leq \varepsilon$. Thus $f_n \to f$ in the norm $\|\cdot\|_\infty$. f is continuous since $f_n \to f$ uniformly.

3. ℓ_∞ is a Banach space. Given Cauchy sequence $\{x^{(m)}\}_{m=1}^{\infty}$, note that $|x_n^{(m)} - x_n^{(k)}| \leq \|x^{(m)} - x^{(k)}\|_\infty$. Therefore, for each n, $\{x_n^{(m)}\}_{m=1}^{\infty}$ is a Cauchy sequence in $\mathbb{C}$, and thus converges. Let x_n be the limit of $\{x_n^{(m)}\}_{m=1}^{\infty}$. We claim that $x^{(m)} \to x = \{x_n\}_{n=1}^{\infty}$, and that $\{x_n\}_{n=1}^{\infty} \in \ell_\infty$. Since $\{x^{(m)}\}_{m=1}^{\infty}$ is Cauchy, for any $\varepsilon > 0$, there exists $M > 0$ such that for any $m, k > M$, $\|x^{(m)} - x^{(k)}\|_\infty < \varepsilon$. Thus for any n, $|x_n^{(m)} - x_n^{(k)}| \leq \varepsilon$. Letting $k \to \infty$, we see that $|x_n^{(m)} - x_n| \leq \varepsilon$ for any n. Therefore, $\left\|x^{(m)} - x\right\| \leq \varepsilon$. Hence $x^{(m)} \to x$ in ℓ_∞. Moreover, $x = x^(m) + (x - x^(m))$. So $x \in \ell_\infty$.

   Since a closed subspace of a Banach space is also a Banach space, we see that C and C_0 are Banach spaces as well.

Theorem 1.1.1 (Completion theorem)**.** Let (X, d) be a metric space, then there exists a complete metric space (Y, p) and a mapping $T : X \to Y$ such that

(1) $T(X)$ is dense in Y.

(2) For any $x, y \in X$, $p(T(x), T(y)) = d(x, y)$.

Moreover, if there is another complete metric space (Y', p') and a mapping $T' : X \to Y'$ satisfying (1) and (2), then there exists a bijection $S : Y \to Y'$ such that for any $x, y \in Y$, $p'(S(x), S(y)) = p(x, y)$.

Remark. Note that we can define the norm $|x| = d(x, 0)$.

Proof. Let $\widetilde{Y}$ be the set of all Cauchy sequences in X. Define a function $\widetilde{p} : \widetilde{Y} \times \widetilde{Y} \to [0, +\infty)$ by

$$\widetilde{p}(\{x_n\}_{n=1}^{\infty}, \{y_n\}_{n=1}^{\infty}) = \lim_{n\to\infty} d(x_n, y_n).$$

Note that this is well-defined. We show that the sequence $\{d(x_n, y_n)\}_{n=1}^{\infty}$ is a Cauchy sequence. By the triangle inequality, we have

$$d(x_n, y_n) \leq d(x_n, x_m) + d(x_m, y_m) + d(y_m, y_n)$$

so $|d(x_n, y_n) - d(x_m, y_m)| \leq d(x_n, x_m) + d(y_n, y_m)$. Since both $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ are Cauchy sequences. For any $\varepsilon/2 > 0$, we can pick N_1 and N_2 such that for any $n, m > N_1$, $d(x_n, x_m) < \varepsilon/2$ and for any $n, m > N_2$, $d(y_n, y_m) < \varepsilon/2$. Let $N = \max\{N_1, N_2\}$. Then for any $n, m > N$, we have $|d(x_n, y_n) - d(x_m, y_m)| < \varepsilon$. Therefore $\{d(x_n, y_n)\}_{n=1}^{\infty}$ is a Cauchy sequence. Since $\mathbb{R}$ is complete, the limit exists.

Then, for any $\{x_n\}_{n=1}^{\infty}, \{y_n\}_{n=1}^{\infty}, \{z_n\}_{n=1}^{\infty} \in \widetilde{Y}$, it is clear that $\widetilde{p}(\{x_n\}_{n=1}^{\infty}, \{z_n\}_{n=1}^{\infty}) \leq \widetilde{p}(\{x_n\}_{n=1}^{\infty}, \{y_n\}_{n=1}^{\infty}) + \widetilde{p}(\{y_n\}_{n=1}^{\infty}, \{z_n\}_{n=1}^{\infty})$.

To make it into a normed space, we need $\tilde{p}(\{x_n\}_{n=1}^{\infty}, \{0\}_{n=1}^{\infty}) = 0 \Leftrightarrow \{x_n\}_{n=1}^{\infty}$ is the zero sequence. We define an equivalence relation $\sim$ on $\widetilde{Y}$ by $\{x_n\}_{n=1}^{\infty} \sim \{y_n\}_{n=1}^{\infty}$ if $d(x_n, y_n) \to 0$ as $n \to \infty$. It is easy to check this is indeed an equivalence relation.

Set $Y = \widetilde{Y}/\sim$, and for $y = [\widetilde{y}]$, $z = [\widetilde{z}]$, set $p(y, z) = \widetilde{p}(\widetilde{y}, \widetilde{z})$. It is clear that this definition is independent to the choice of representative. It makes (Y, p) into a metric space. Define $T : X \to Y$ by $T(x) = \{x_n\}_{n=1}^{\infty}$ where $x_n = x$ for all n. Then, for any $x, y \in X$, we have $p(T(x), T(y)) = d(x, y)$. Also, $T(X)$ is dense in Y. Indeed, let $y \in Y$. Let $\{y_n\}_{n=1}^{\infty}$ be a Cauchy sequence in X that represents y. Set $\overline{x} = Ty_n$, i.e., $\overline{x_n} = \{y_n, y_n, \ldots\}$. Then

$$p(\overline{x_n}, y) = \lim_{m\to\infty} d(y_n, y_m) \xrightarrow{n\to\infty} 0.$$

Next, we show completeness of (Y, p). Let $\{y_n\}_{n=1}^{\infty} \subset Y$ be a Cauchy sequence. Since $T(X)$ is dense in Y, for any $n \in \mathbb{N}$, we can choose $x_n \in X$ such that $p(Tx_n, y_n) < 1/n$. Then it is not hard to see that $\{Tx_n\}_{n=1}^{\infty}$ is a Cauchy sequence in Y. Therefore, $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence in X. Set $y = [\{x_n\}_{n=1}^{\infty}]$. Then $Tx_n \to y$. So $y_n \to y$ as well.

Finally, we show the uniqueness. Let (Y', p') be another complete metric space and $T' : X \to Y'$ be a mapping such that $T'(X)$ is dense in Y' and $p'(T'x, T'y) = d(x, y)$. Define $S : T(X) \to T'(X)$ by $Tx = [(x, x, \ldots)] \mapsto T'x$. Notice that since T is injective, this is well-defined. Moreover, for any $y, z \in T(X)$, $p'(Sy, Sz) = p(y, z)$. Now, since $T(X)$ is dense in Y and $T'(X)$ is dense in Y', we can extend S by continuity. More precisely, given $y \in Y$, choose $Tx_n \to y$. Then $T'x_n \to y'$ for some $y' \in Y'$. Define $Sy = y'$. It is easy to check that this is independent to the choice of sequence. We see that his is a bijection, since S is an isometry. Uniqueness comes from continuity.

∎

Lecture 2: Sobolev Spaces and Linear Operators

We start with an application of the completion theorem.

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Example. Let $X = C([0,1])$ be the space of continuous functions on $[0,1]$, where the norm is given by

$$\|f\| = \left(\int_0^1 |f(x)|^p dx\right)^{1/p}, \quad p \in [1, \infty)$$

Claim. completion$(X) \cong L_p([0,1])$.

Proof. Its enough to check that X is dense in $L_p([0,1])$, and $L_p([0,1])$ is complete. These are standard results in Math 597. ■

1.2 Sobolev Space, an informal introduction

Definition 1.2.1. Let $X = C^m([a,b])$ be the space of m-times continuously differentiable functions on $[a,b]$. Let $p \in [1, \infty)$, and define the norm

$$\|f\|_{W_p^m((a,b))} = \left(\sum_{j=0}^{m} \int_a^b |f^{(j)}(x)|^p dx\right)^{1/p}$$

Let $W_p^m((a,b))$ be the completion of X.

Remark. This is a nonstandard definition of Sobolev space, but it is equivalent to the standard definition. But this definition is more intuitive.

Remark. We can extend this definition to higher dimensions. Specifically, let $\Omega \subseteq \mathbb{R}^n$ be an open set, and denote $X = C^m(\Omega)$ the space of m-times continuously differentiable functions on $\Omega \subset \mathbb{R}^n$. Use $\alpha = (\alpha_1, \ldots, \alpha_n)$ to denote a multi-index, and define the norm

$$|f| = \left(\sum_{|\alpha| \leq m} \int_\Omega |D^\alpha f(x)|^p dx\right)^{1/p}$$

Consider $m = 1$. Our goal is to understand $W_p^1((a,b))$.

Theorem 1.2.1. Given $F \in W_p^1((a,b))$, there exists a pair (f, g), where $f \in C((a,b)) \cap L_p((a,b))$, $g \in L_p((a,b))$, such that f represents the same L_p class to F, and for any $x, y \in (a,b)$, with $x < y$,

$$f(y) - f(x) = \int_x^y g(t)dt \tag{*}$$

Such g is called the *generalized derivative*.

Proof. Let $\{f_n\}_{n=1}^\infty$ be a Cauchy sequence in $C^1((a,b))$ converging to F in $W_p^1((a,b))$. Since $\|f_n\| = \left(\int_a^b |f_n(x)|^p dx + \int_a^b |f_n'(x)|^p dx\right)^{1/p} = \left(\|f_n\|_p^p + \|f_n'\|_p^p\right)^{\frac{1}{p}}$, we see that both $\{f_n\}_{n=1}^\infty$ and $\{f_n'\}_{n=1}^\infty$ are Cauchy sequences in $L_p(a,b)$. Let

$$f_n \to f \quad \text{and} \quad f_n' \to g \quad \text{in } L_p(a,b)$$

Claim. $f_n \to f$ uniformly on (a,b). So that in particular, $f \in C((a,b))$.

Proof. Let $x < y$, $x, y \in (a,b)$. Since $f_n \in C^1([a,b])$, we have, by Fundamental Theorem of Calculus,

$$f_n(y) - f_n(x) = \int_x^y f_n'(t)dt$$

By Hölder's inequality, we have

$$\begin{aligned} |f_n(y) - f_n(x)| &\leq (y-x)^{1/q} \left(\int_x^y |f_n'(t)|^p dt \right)^{1/p} \\ &\leq (b-a)^{1/q} \|f_n\|_{W_p^1((a,b))} \end{aligned}$$

Since $\{f_n\}_{n=1}^\infty$ is a Cauchy sequence in $W_p^1((a,b))$, $\|f_n\|_{W_p^1((a,b))}$ is bounded. Therefore, $|f_n(y) - f_n(x)|$ is uniformly bounded on (a,b).

Moreover, by the triangle inequality, $|f_n(y)| \leq |f_n(x)| + |f_n(y) - f_n(x)| \leq |f_n(x)| + C_{a,b} \|f_n\|_{W_p^1((a,b))}$ for some constant $C_{a,b}$. Since it holds for all $x \in (a,b)$, we get

$$\begin{aligned} |f_n(y)| &\leq \frac{1}{b-a} \int_a^b |f_n(x)|\, dx + C_{a,b} \|f_n'\|_{W_p^1((a,b))} \\ &\overset{\text{Hölder}}{\leq} C'_{a,b} \|f_n\|_p + C_{a,b} \|f_n'\|_{W_p^1((a,b))} \\ &\leq \widetilde{C_{a,b}} \|f_n\|_{W_p^1((a,b))} \end{aligned}$$

Therefore, $|f_n(y) - f_m(y)| \leq \widetilde{C_{a,b}} \|f_n - f_m\|_{W_p^1((a,b))} \to 0$ since $\{f_n\}_{n=1}^\infty$ is a Cauchy sequence in $W_p^1((a,b))$. This shows that $\{f_n\}_{n=1}^\infty$ is uniformly Cauchy, hence converges uniformly to f. In particular, $f \in C((a,b))$. Moreover, $f \in \mathrm{Hol}_{1/q}((a,b))$. ∎

Then, we use

$$f_n(y) - f_n(x) = \int_x^y f_n'(t)dt$$

We already have $f_n(y) \to f(y)$ and $f_n(x) \to f(x)$, in a uniform sense, and $f_n' \to g$ in L_p, thus in L_1. Thus $\int_x^y f_n'(t)dt \to \int_x^y g(t)dt$. Hence, for any $x, y \in (a,b)$, $x < y$ we have

$$f(y) - f(x) = \int_x^y g(t)dt \tag{*}$$

∎

There is a useful application of it to the change of variables.

Note. Let $\varphi \in C^\infty((a,b))$ be such that $\mathrm{supp}(\varphi) \subset (a,b)$. Therefore the change of variables formula gives

$$\int_a^b f_n(x)\varphi'(x)dx = -\int_a^b f_n'(x)\varphi(x)dx$$

By Hölder's inequality, we have

$$\left| \int_a^b (f_n - f)(x)\varphi'(x)dx \right| \leq \|f_n - f\|_p \|\varphi'\|_q \to 0$$

The RHS is similar, with $\left|\int_a^b (f_n' - g)(x)\varphi(x)dx\right| \to 0$. Therefore, we have

$$\int_a^b f(x)\varphi'(x)dx = -\int_a^b g(x)\varphi(x)dx$$

These are functions in Sobolev space that are not differentiable in the usual sense. Here is an example

Example. Take $(a,b) = (-1,1)$ and then $(|x|, \mathrm{sgn}(x)) \in W_1^1((-1,1))$.

1.3 Linear Operators

Definition 1.3.1. A *linear operator* $T : X \to Y$ between Banach spaces X and Y is called *bounded* if

$$\|T\| = \|T : X \to Y\| = \sup\{\|Tx\|_Y : \|x\|_X \leq 1\} < \infty$$

Here are two examples on why we are interested in linear operators.

- Solve the Laplace equation $\Delta f = g$. We can think of Δ as a linear operator, but in general it is not bounded. The other way to think about this problem is to give a g, and find an f such that $\Delta f = g$. This is also a linear operator.

- Fourier transform: The Fourier transform of f is

$$\hat{f}(t) = \int_{\mathbb{R}} \exp(2\pi i t x) f(x) dx$$

  which is a bounded linear map sends $L_1(\mathbb{R})$ to $L_\infty(\mathbb{R})$.

Lemma 1.3.1. Let $T : X \to Y$ be a linear operator between Banach spaces. The following are equivalent:

1. T is continuous.
2. T is continuous at 0.
3. T is bounded.

Proof. "1 $\Rightarrow$ 2" is trivial.
"2 $\Rightarrow$ 3": Let T be continuous at 0. Then there exists $\delta > 0$, such that for any $x \in X$, $\|x\|_X < \delta$ implies $\|Tx\|_Y < 1$. Then for any $y \in X$ with $\|y\|_X \leq 1$, we have $\|Ty\|_Y = \delta^{-1} \|T(\delta y)\| < \delta^{-1}$.
"3 $\Rightarrow$ 1": Use $\|T(x - y)\|_Y \leq \|T\| \cdot \|x - y\|_X$. ■

Lemma 1.3.2. Let $L(X, Y)$ be the space of bounded linear operators from X to Y, equipped with the norm $\|\cdot : X \to Y\|$ defined as above. Then $L(X, Y)$ is a Banach space.

Proof. HW exercise. ■

1.4 Baire's category theorem

Theorem 1.4.1. Let (X, d) be a complete metric space, and let $\{E_n\}_{n=1}^\infty$ be a sequence of nowhere dense subsets of X. Then $X \setminus \bigcup_{n=1}^\infty E_n$ is dense in X.

Proof. It suffices to show that $X \neq \bigcup_{n=1}^\infty \overline{E_n}$. Note that given $x \in X$, and arbitrary closed ball $\overline{B(x,r)}$, we can consider $\{\overline{E_n} \cap \overline{B(x,r)}\}_{n=1}^\infty$. We can then replace X by $\overline{B(x,r)}$, which is still a complete metric space with the subspace metric. Then we see that $\overline{E_n}$ can't cover $\overline{B(x,r)}$.

Assume that $X = \bigcup_{n=1}^\infty \overline{E_n}$. We can choose $x_1 \in X$, $r_1 > 0$ such that $\overline{B(x_1, r_1)} \cap \overline{E_1} = \varnothing$. We construct $\{x_n\}_{n=1}^\infty$ and $\{r_n\}_{n=1}^\infty$ inductively. Assume that we have constructed $x_1, \ldots, x_n$ and $r_1, \ldots, r_n$ such that $\overline{B(x_n, r_n)} \cap (\bigcup_{k=1}^n \overline{E_k}) = \varnothing$, then there exists a ball $\overline{B(x_{n+1}, r_{n+1})} \subset \overline{B(x_n, r_n)}$ such that $\overline{B(x_{n+1}, r_{n+1})} \cap \overline{E_{n+1}} = \varnothing$. WLOG, assume that $r_{n+1} < r_n/2$. Then $\{x_n\}_{n=1}^\infty$ is a Cauchy sequence, and since X is complete, there exists $x \in X$ such that $x_n \to x$. Moreover, $x \in \overline{B(x_n, r_n)}$ for all n, since each ball is closed. We thus see that $x \in \bigcap_{n=1}^\infty \overline{B(x_n, r_n)}$, which is a contradiction. ■

Theorem 1.4.2 (Banach-Steinhaus theorem). Let $\{T_n : X \to Y_n\}_{n=1}^\infty$ be a sequence of bounded linear operators between Banach spaces. If for any $x \in X$, $\sup_{n \in \mathbb{N}} \|T_n x\|_{Y_n} < \infty$, then $\sup_{n \in \mathbb{N}} \|T_n : X \to Y_n\| < \infty$.

Proof. Let $E_N = \{x \in X : \forall n \in N, \|T_n x\|_{Y_n} \leq N\}$. Then $X = \bigcup_{n=1}^{\infty} E_n$. By Theorem 1.4.1, there exists $N \in \mathbb{N}$, $x \in X$, and $r > 0$ such that $\overline{B(x,r)} \subset E_N$. Then for any $n \in \mathbb{N}$, and any $y \in X$ for which $\|y\|_X \leq r$, one has $\|T_n y\|_{Y_n} \leq N$. Therefore, $\|T_n(x,y)\|_{Y_n} \leq 2N$. Then, for any $z \in X$ and any n, we have $\|T_n z\|_{T_n} \leq 2N$ when $\|z\|_X \leq r$. Therefore, $\|T_n\| \leq 2N/r$ which is a constant. ■

Appendix

Appendix A

Finite and integral homomorphisms

A.1 Definitions

Let $\varphi : R \to S$ be a ring homomorphism. One say that φ is of *finite type* if S is a finitely generated R-algebra. One say that φ is *finite* if S is a finitely generated R-module. One say that φ is *integral* if every element of S is integral over R, i.e., for every $s \in S$, there exists a monic polynomial $f(x) \in R[x]$ such that $f(s) = 0$.

Remark. It is clear that if φ is finite, then it is of finite type. The converse is not true. For example, take the inclusion $R \hookrightarrow R[x]$. Then $R[x]$ is a finitely generated R-algebra, but it is not a finitely generated R-module, for finitely many polynomials only generate bounded degree.

Remark. If φ is of finite type and integral, then it is finite. Let $y_1, \ldots y_r$ generates S as an R-algebra. Since φ is integral, we can write

$$y_i^{d_i} + a_{i,1}y_i^{d_i-1} + \ldots + a_{i,d_i} = 0$$

Then it is easy to see that

$$\{y_1^{a_1} \cdots y_r^{a_r} \mid 0 \leq a_i \leq d_i - 1\}$$

generates S as an R-module.

Proposition A.1.1. If φ is finite, then it is integral.

Proof. Suppose that $b_1, \ldots b_n$ generate S as an R-module, then for every $y \in S$, write, for each $1 \leq i \leq n$,

$$yb_i = \sum_{j=1}^{n} a_{i,j}b_j \quad \text{for some } a_{i,j} \in R$$

Let A be the matrix $(a_{i,j})_{1\leq i,j\leq n}$, then we have

$$(yI - A) \cdot \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0$$

Multiplying by the adjoint matrix of $(yI - A)$, we see that if $D = \det(yI - A)$, then $Db_i = 0$ for every $1 \leq i \leq n$. This imples $D \cdot 1_S = 0$. However, it is clear that we can write

$$D = y^n + c_1y^{n-1} + \ldots + c_n \quad \text{for some } c_i \in R$$

We see that y is integral over R. ∎

Remark. It follows from Proposition A.1.1 and the previous remark that if φ is of finite type, then it is finite if and only if it is integral.

A.2 Easy properties

Proposition A.2.1. Let $\varphi : R \to S$ be an injective, integral homomorphism of integral domains, then R is a field if and only if S is a field.

Proof. Suppose that R is a field, and let $u \in S \setminus \{0\}$. Since φ is integral, we can write

$$u^n + a_1 u^{n-1} + \ldots + a_n = 0 \quad \text{for some } a_i \in R$$

We see that $u(u^{n-1} + a_1 u^{n-2} + \ldots + a_{n-1}) = -a_n$. Since $u \neq 0$, $a_n \neq 0$. Since R is a field, a_n is invertible in R, we see that $uv = 1$, where

$$v = (-a_n)^{-1}(u^{n-1} + a_1 u^{n-2} + \ldots + a_{n-1}) \in S$$

Since it holds for all $u \in S \setminus \{0\}$, we see that S is a field. Conversely, suppose that S is a field, and let $a \in R \setminus \{0\}$. Let $b = a^{-1} \in S$, we want so show that $b \in R$. Since φ is integral, we can write

$$b^n + \alpha_1 b^{n-1} + \ldots + \alpha_n = 0 \quad \text{for some } \alpha_i \in R$$

but then $b = -\alpha_1 - \alpha_2 a - \ldots - \alpha_n a^{n-1} \in R$. Since it holds for all $a \in R \setminus \{0\}$, we see that R is a field. ■

Proposition A.2.2. Given a ring homomorphism $\varphi : R \to S$, the subset

$$S' \coloneqq \{y \in S \mid y \text{ is integral over } R\}$$

is a subring of S. This is the *integral closure* of R in S.

Proof. It is easy to see that $1_s \in S'$. It remains to show that if $y_1, y_2 \in S'$, then $y_1 - y_2, y_1 y_2 \in S'$. Since y_1, y_2 are integral over R, we know that the ring $R[y_1, y_2]$ is finite over R, hence integral over R by Proposition A.1.1. This implies that $y_1 - y_2, y_1 y_2 \in S'$. ■

Proposition A.2.3. Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms. If φ and ψ are of finite type (respectively finite, integral), then so is $\psi \circ \varphi$.

Proof. The assertion is straightfoward for finite and finite type morphisms. Suppose now that both φ and ψ are integral. Let $u \in T$, then u is integral over S, so we can write

$$u^n + b_1 u^{n-1} + \ldots + b_n = 0 \quad \text{for some } b_i \in S$$

Since $b_1, \ldots b_n$ are integral over R, the ring $R' \coloneqq R[b_1, \ldots b_n]$ is finite over R. Moreover, $R'[u]$ is finite over R' since u is integral over R'. Hence $R'[u]$ is finite over R, hence integral over R by Proposition A.1.1. ■